Indian Institute of Technology Jammu
Department of Computer Science and Engineering

Mid-Term Exam, Sep 20, 2025

Subject: Computers and Programming | Course Code: CS-1-01(MO)

Time: 90 Minutes | Full Marks: 30

Name: [redacted]

Student ID: 2025UMT0214

Instructions

This exam has 2 printed page/s. Each Question carries two marks. Answer all the questions. The question number **15** is mandatory. **Illogical or incoherent answers are worse than wrong answers or even no answers and may fetch negative credit.** You may not use any computing or communication device during the exam. **The exam will be closed notes. No clarification during the exam.**

Questions

1. Define the terms **Computer**, **Machine Language**, **Assembly Language**, and **High-Level Language** with suitable examples.

2. Perform the addition of two octal numbers $(347)_8$ and $(265)_8$ using the **8's complement representation** in a 3-digit number system. Show all the steps.

3. Subtract the Hexadecimal numbers $(412)_16$ and (7[illegible]6)$_16$ using **16's complement method** in a 3-digit number system. Show all the steps.

4. Represent the real number $+10.75$ in **IEEE 754 half precision** format. Show all the steps.

5. Represent the real number -27.625 in **IEEE 754 single precision** format. Show all the steps.

6. Explain the different **steps of C program compilation** (from source code to executable).

7. Differentiate between a **Linker** and a **Loader** in C programming.

8. Differentiate between a **Object** file and a **Executable** file in C programming.

9. Identify the **valid and invalid identifiers** from the following list and give reasons: _sum, 2ndValue, float, my_variable, total_marks, $price.

10. Evaluate the following expression (value of a) step by step using operator precedence and associativity rules:

$$a = 25 + 12 \times 4 - 36/\,6\,\%\,2;$$

11. Evaluate the expression (value of b) with proper precedence and associativity:

$$b = (35\%6) \times 4 + 18/(3+3)\%4/3;$$

12. Find and correct the errors in the following C program:

```
#include <stdio.h>
int main() {
    int a, b;
    a = 10; b = 0
    printf("Result = d", a/b);
    return 0;
}
```

13. Find the errors in this C program and rewrite the correct version:

```
#include <stdio.h>

int main() {
    int _a = 10, $b = 20;
    int $p;
    $p = (a > b) ? _a : $b
    print("The value of p is: %c\n", max);
    return 0;
}
```

14. Write steps to compute the **square root of a number** in:

(a) Imperative style

(b) Declarative style

15. Answer the following questions:

(a) Write the name of the Teaching Assistant (TA) assigned to you.

(b) How frequently do you discuss your problems with the assigned TA?

------------ END of Exam ------------

CS-1-02(MO) : Data Structures and Algorithms
Mid-Sem

February 18, 2026

Total Marks : 26 | Duration: 2 hrs. | Maximum Marks : 25

Instructions:

- Read the questions carefully and make sure that you understand them before starting to write your code.
- Mobile phones and smart devices are strictly prohibited in the exam hall. Do not bring them with you.
- Cheating in any form will result in zero marks for the exam. Additionally, a penalty of minus 5 marks will be given.
- All the questions are self-explanatory, however, you are allowed to ask any doubt.

Questions:

1. Attempt any one: (a) or (b)

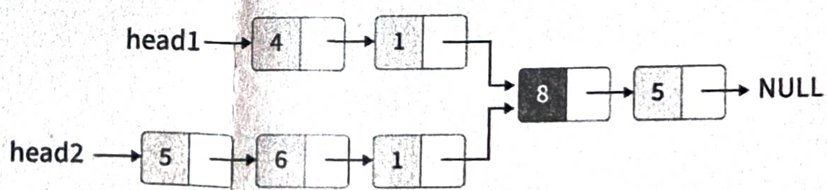
   (a) Let $T_1(n)$, $T_2(n)$ and $f(n)$ be three positive functions which are defined as follows:

   i. $T_1(n) = O(f(n))$
   ii. $T_2(n) = O(f(n))$

   Write true or false with proper justification. [1.5+1.5+1.5+1.5]

   i. $T_1(n) + T_2(n) = O(f(n))$
   ii. $T_1(n) = O(T_2(n))$
   iii. $T_1(n) = \Omega(T_2(n))$
   iv. $T_1(n) = \Theta(T_2(n))$

   (b) You are given the heads of two singly linked lists. The two lists may have different lengths and contain distinct nodes initially. However, at some point, the two lists merge and share the same sequence of nodes from the intersection point onward.
   Write a C program/C++ program/pseudocode to find and return the node at which the two linked lists first intersect. [6]



   Output: Node with a stored data value to be 8.

2. Write a C program/C++ program/pseudocode to reverse a Singly Linked List. What is the time complexity? [5]

   (Time complexity is a measure of how the running time of an algorithm grows as a function of the input size n.)

3. Write a C program/C++ program/pseudocode to find and remove the Nth node from the end of a Singly Linked List.

   Your program should: [5]

   (a) Take an integer N as input, representing the position of the node from the end to be removed.

   (b) Remove the Nth node from the end of the linked list.

   (c) Print the linked list after the removal.

4. Write a C program/C++ program/pseudocode to remove the largest element in a given stack S and construct S again with elements in the same order but without the largest element. You may use additional data structure but not allowed to use stack or queue directly from STL in C++. [5]

5. Convert the following infix to postfix using stack. Show intermediate stack state.

   ```
   ((d-e)*(b+a))^c^(f/(g-h)*i)
   ```

   The order of precendence from higher to lower is

   (a) `(,)`- Parentheses

   (b) `^` - Exponentiation (Right to Left associativity)

   (c) `*,/` - Multiplication and Division (Left to Right associativity)

   (d) `+,-` - Addition and Subtraction (Left to Right associativity)

   What is the time and space complexity? [3+1+1]

   (An intermediate stack state is the contents of the stack at any point during the execution of an algorithm.

   Space complexity is a measure of how much memory an algorithm requires as a function of input size n. To measure space complexity, we take into account of extra memory used by the algorithm other than the memory required for input and output.)

CS-1-02(MO) : Data Structures and Algorithms
End-Sem

April 21, 2026

Total Marks : 40 | Duration: 3 hrs. | Maximum Marks : 40

Instructions:

- Read the questions carefully and make sure that you understand them before starting to write your code.
- Mobile phones and smart devices are strictly prohibited in the exam hall. Do not bring them with you.
- Cheating in any form will result in zero marks for the exam. Additionally, a penalty of minus 5 marks will be given.
- All the questions are self-explanatory, however, you are allowed to ask any doubt.

Questions:

1. Solve any one - (a) or (b) [5]

   (a) Insert 7 into a min-heap tree with elements 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200. What is the time complexity for this case.,

   (b) A k-balanced binary search tree (called a **VISA Tree**) is defined as a BST in which, for every node:

   $$\text{Balance Factor} = \text{height(left subtree)} - \text{height(right subtree)}$$

   The balance factor is allowed in the range $[-k, +k]$. Here, assume $k = 2$.

   (a) Write the pseudo/C++/C code to insert a node in a VISA tree. Also, insert the following keys into an empty k-balanced tree:

   $$10,\ 20,\ 30,\ 40,\ 50,\ 60,\ 70$$

2. Write the pseudo/C/C++ code for Quicksort. What is the best and worst case time complexity? [5]

3. Solve any one - (a) or (b) [5]

   (a) You are given $\log n$ sorted subarrays each of size $n/\log n$. Output is a single sorted array of n elements. Find time complexity.

   (b) Given a preorder and in order sequences of a binary tree as $D, E, B, G, A, F, C, I, H$ and $E, G, A, B, D, C, H, I, F$ respectively. Construct the tree and find the postorder tree traversal sequence.

4. Solve any one - (a) or (b) [5]

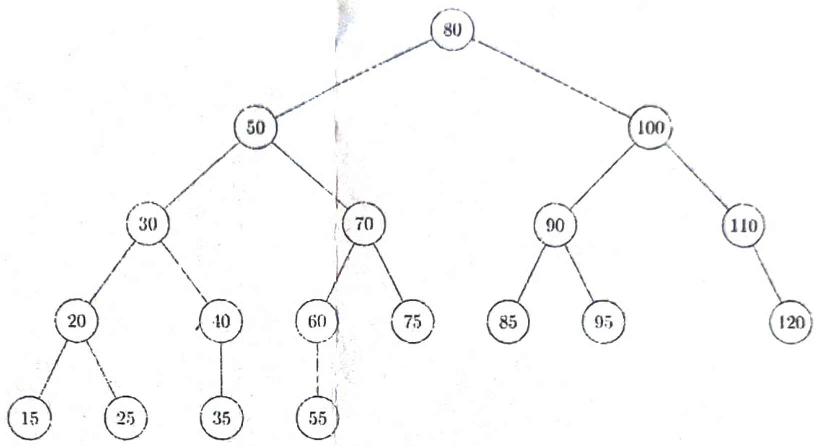
   (a) Solve the recurrence relation:

   $$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n-1) * n & \text{if } n > 1 \end{cases}$$

(b) Implement a descending priority queue using unsorted arrays (Input = 25,77, 92, 9, 45, 17) . What is the worst time taken for one Enqueue and one Dequeue operation?

A descending priority queue is a specialized queue data structure where elements with the highest numerical value are assigned the highest priority and are removed (dequeued) first.

5. Delete 80 (a node with two children) from the given AVL tree. [5]



Write the total time complexity with breakup for each action.

6. A Tri-Search Tree (TST) is defined as:

- Left child → keys less than node value
- Middle child → keys equal to node value
- Right child → keys greater than node value

(a) Write the pseudo/C++/C code for insertion in TST. Also, insert the following keys into an empty TST and draw the final tree: [5]

10, 5, 15, 10, 7, 10, 3, 15, 20, 5, 7, 2

(b) Write the pseudo/C++/C code for deletion in TST. Also, delete the following keys from the constructed tree: [5]

10, 5, 15

7. You are given a singly linked list where each node stores a string. [5]

"banana" → "apple" → "cherry" → "avocado" → "blueberry"

Write pseudo / C++ / C code to rearrange the linked list such that it is sorted based on the **first character** of each string (lexicographically). The output for the above example is:

"apple" → "avocado" → "banana" → "blueberry" → "cherry"

- Do not create a new list; modify the existing list.
- If you make assumptions about ASCII values, clearly state them.